

Machine-Learning-Based Variance Reduction for Monte Carlo Pricing and Risk Computation in Financial Markets

Topic Description

Monte Carlo methods are a central tool in market finance for pricing and risk computation under realistic stochastic models, as introduced in courses on stochastic processes, Monte Carlo methods, and advanced statistics. However, their intrinsic slow convergence leads to high variance, noisy prices, and unstable Greeks, which severely limits their usability in front-office and asset management environments. This project proposes to address this structural limitation by reformulating variance reduction as a high-dimensional functional optimisation problem. Machine learning techniques are used to learn optimal control variates and importance sampling schemes directly from simulated data, while preserving martingale and no-arbitrage constraints. The objective is to achieve faster convergence, improved numerical stability, and computational efficiency compatible with real-time pricing, hedging, and risk management applications.

Selected Recent References

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